

Question 1

Given an array of strings `arr[]` of length `n`, where every string representing a non-negative integers, the task is to arrange them in a manner such that after concatenating them in order, it results in the largest possible number. Since the result may be very large, return it as a string.

Examples:

Input: `n = 5, arr[] = {"3", "30", "34", "5", "9"}`

Output: "9534330"

Explanation: Given numbers are {"3", "30", "34", "5", "9"}, the arrangement "9534330" gives the largest value.

Input: `n = 4, arr[] = {"54", "546", "548", "60"}`

Output: "6054854654"

Explanation: Given numbers are {"54", "546", "548", "60"}, the arrangement "6054854654" gives the largest value.

Question 2

Given an array of strings `words` and an integer `k`, return the `k` most frequent strings.

Return the answer sorted by the frequency from highest to lowest. Sort the words with the same frequency by their lexicographical order.

A **lexicographic order** is an arrangement of characters, words, or numbers in alphabetical order, that is, the letters are sorted from A-Z.

This is also known as **dictionary order** because it is similar to searching for a particular word in an actual dictionary. We start by searching for the section containing the words starting with the first letter of the desired word. From that section, we search for the words that contain the second letter of the desired word, and so on.

Example 1:

Input: words = ["i","love","leetcode","i","love","coding"], k = 2

Output: ["i","love"]

Explanation: "i" and "love" are the two most frequent words.

Note that "i" comes before "love" due to a lower alphabetical order.

Example 2:

Input: words = ["the","day","is","sunny","the","the","the","sunny","is","is"], k = 4

Output: ["the","is","sunny","day"]

Explanation: "the", "is", "sunny" and "day" are the four most frequent words, with the number of occurrence being 4, 3, 2 and 1 respectively.

Constraints:

- $1 \leq \text{words.length} \leq 500$
- $1 \leq \text{words}[i].\text{length} \leq 10$
- `words[i]` consists of lowercase English letters.
- `k` is in the range $[1, \text{The number of unique words}[i]]$

Solution:

```
def topKFrequent(words: List[str], k: int) -> List[str]:
```

Question 3

From below sample JSON,

1. Filter all users whose age is below 50
2. Filter all users whose gender is Female and whose age is above 30

Sample JSON:

```
[
  {
    "id":1,
    "first_name":"Hershel",
    "last_name":"Andriss",
    "email":"handriss0@behance.net",
    "gender":"Male",
    "age":68
  },
  {
    "id":2,
    "first_name":"Goldia",
    "last_name":"Sunman",
    "email":"gsunman1@homestead.com",
    "gender":"Female",
    "age":36
  },
  {
    "id":3,
    "first_name":"Tobie",
    "last_name":"Dolan",
    "email":"tdolan2@msu.edu",
    "gender":"Male",
    "age":20
  },
  {
    "id":4,
    "first_name":"Chaunce",
    "last_name":"Fulep",
    "email":"cfulep3@nature.com",
```

```
"gender":"Genderfluid",
"age":29
},
{
  "id":5,
  "first_name":"Edlin",
  "last_name":"Martland",
  "email":"emartland4@altervista.org",
  "gender":"Male",
  "age":33
},
{
  "id":6,
  "first_name":"Nappy",
  "last_name":"Lanfear",
  "email":"nlanfear5@unblog.fr",
  "gender":"Male",
  "age":46
},
{
  "id":7,
  "first_name":"Maggie",
  "last_name":"Kilmary",
  "email":"mkilmary6@zimbio.com",
  "gender":"Female",
  "age":58
},
{
  "id":8,
  "first_name":"Kally",
  "last_name":"Marling",
  "email":"kmarling7@opensource.org",
  "gender":"Female",
  "age":26
},
{
  "id":9,
  "first_name":"Lucinda",
  "last_name":"Camacho",
  "email":"lcamacho8@whitehouse.gov",
```

```
"gender":"Non-binary",
"age":24
},
{
  "id":10,
  "first_name":"Maxine",
  "last_name":"Sawell",
  "email":"msawell9@bing.com",
  "gender":"Female",
  "age":61
},
{
  "id":11,
  "first_name":"Devon",
  "last_name":"Hub",
  "email":"dhuba@si.edu",
  "gender":"Female",
  "age":37
},
{
  "id":12,
  "first_name":"Lenette",
  "last_name":"Garside",
  "email":"lgarsideb@hhs.gov",
  "gender":"Female",
  "age":21
},
{
  "id":13,
  "first_name":"Trip",
  "last_name":"Leming",
  "email":"tlemingc@mysql.com",
  "gender":"Male",
  "age":49
},
{
  "id":14,
  "first_name":"Dill",
  "last_name":"Robak",
  "email":"drobakd@tinypic.com",
```

```
"gender":"Male",
"age":43
},
{
  "id":15,
  "first_name":"Nedi",
  "last_name":"Levins",
  "email":"nlevinse@cyberchimps.com",
  "gender":"Polygender",
  "age":68
},
{
  "id":16,
  "first_name":"Horacio",
  "last_name":"Mustin",
  "email":"hmustinf@amazon.de",
  "gender":"Genderqueer",
  "age":45
},
{
  "id":17,
  "first_name":"Dalis",
  "last_name":"Lambard",
  "email":"dlambardg@pen.io",
  "gender":"Male",
  "age":32
},
{
  "id":18,
  "first_name":"Lem",
  "last_name":"Abercromby",
  "email":"labercrombyh@admin.ch",
  "gender":"Male",
  "age":34
},
{
  "id":19,
  "first_name":"Nikkie",
  "last_name":"Dieton",
  "email":"ndietoni@hugedomains.com",
```

```
    "gender": "Female",
    "age": 48
  },
  {
    "id": 20,
    "first_name": "Leontine",
    "last_name": "Hockell",
    "email": "lhockellj@etsy.com",
    "gender": "Female",
    "age": 74
  }
]
```